METRIC
DOD-E-24636(SH)
16 July 1984

MILITARY SPECIFICATION

ELECTRODE, pH, COMBINATION SEALED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a combination pH and reference electrode, which is used in conjunction with a digital indicating pH meter to measure the pH of boiler water.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. Unless otherwise specified, the following specifications of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6630

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(Copies of specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM

D 1293 - Standard Test Methods for pH of Water. (DoD adopted)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 General. The combination electrode and its connecting cable and connectors shall be compatible for use with a Seasorex model S200C digital pH meter, or equal, and shall be used for pH measurements in boiler water samples aboard Naval ships at sea.

3.3 Materials. Materials shall be in accordance with the requirements specified as follows:

3.3.1 Outer body. The protective outer body of the electrode shall consist of a rigid, low actinic transparent plastic material.

3.3.2 Protective end cap closure. The electrode shall be provided with a removable plastic protective end cap closure, with vent hole. The end cap shall contain a soft inner absorbent liner saturated with a three molar potassium chloride solution. The end cap shall be provided with a strip of removable adhesive tape covering the vent hole, in order to prevent loss of electrolyte from the electrode, and to prevent evaporation of the potassium chloride solution in the absorbent liner during shipment and storage.

3.3.3 Electrode glass sensing element. The glass sensing element shall consist of a glass formulation having the physical, chemical and electrical properties that do not interfere with accurate pH determination.

3.3.4 Electrode reference cell. The electrode reference cell shall consist of silver-silver chloride combined with a low viscosity gel of saturated potassium chloride-silver chloride.

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3.3.5 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3.6 Mercury containing components. Electrode components shall not contain mercury in any form.

3.4 Construction.

3.4.1 Rigid outer body. The electrode shall have a rigid plastic outer body, which shall extend at least 1/8-inch below the bottom of the glass sensing element, in order to prevent accidental breakage of the glass, and to enable the electrode to be used as a stirring rod for liquid samples in either glass or plastic containers.

3.4.2 Permanent seal. The combination electrode shall be permanently sealed and non-refillable.

3.4.3 Dimensions and weight. The electrode, less connecting cable, shall be 13 to 18 centimeters (cm) in length, and 1.27 cm in outside diameter, plus or minus 0.25 cm. The complete electrode assembly (electrode, cable and connectors) shall not exceed 60 grams in weight.

3.4.4 Connecting cable and connectors. The electrode connecting cable shall be 76 to 122 cm in length, shall be insulated and shielded and shall be provided with:

(a) a U.S. standard type plug for the sensing element and,
(b) a pin tip connector of outside diameter 0.2 millimeter (mm) $\pm$ 1.01 mm for the reference element.

3.5 Performance.

3.5.1 Range of operation. The combination pH and reference electrode shall operate over the range 0.00 to 13.99 pH units from 0 degrees Celsius (°C) to 100°C.

3.5.2 Precision and accuracy. When the combination electrode is tested in accordance with 4.6.1, 4.6.1.1 and 4.6.1.2, the precision and accuracy criteria shall be plus or minus 0.05 pH unit.

3.5.3 Response time. For samples in equilibrium with the ambient temperature, the attainment of a stable pH reading shall require no longer than 1 minute from time of immersion of the combination electrode in the sample. For samples at temperatures other than ambient, no longer than 5 minutes shall be required for the pH determination. A meter reading that does not vary by more than plus or minus 0.02 pH unit in 30 seconds shall constitute a stable pH reading.

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3.5.4 Use nation electrode shall have a use life of 6 months or longe

3.5.5 Iso The combination electrode shall have its isopotential point a

3.5.6 Eff:celeration. The accuracy and precision criteria of 3.5.2 shall .ectrode is subjected to an angular acceleration created by a ro: to 60 degrees from normal and back to normal position in a pids. The roll shall be repeated about an axis 90 degrees remo

3.6 Identis. Identification markings on the electrode shall show the date of manufacture, model number, serial number, nationa:ontractor and name of the manufacturer. Identification labels, types A, B or C of MIL-P-15024 and MIL-P-15024/5.

3.7 Instrie contractor shall pack with each electrode, instructions fo:e, and storage of the electrode.

3.8 Workmes shall be free from interior or exterior foreign materia:ches or loose particles which may interfere with normal ope:ctrode. Surfaces shall be clean and free from marks.

4. QUALITSIONS

4.1 Respopection. Unless otherwise specified in the contract or purcontractor is responsible for the performance of all inspectics specified herein. Except as otherwise specified in thechase order, the contractor may use his own or any other facilir the performance of the inspection requirements specified hereiroved by the Government. The Government reserves the right to perinspections set forth in the specification where such inspectionssary to assure supplies and services conform to prescribed re

4.2 Classiections. The inspection requirements specified herein are class:

(a) Ipection (see 4.3).
(b) Ce inspection (see 4.4).

4.3 First on. A first article inspection shall be performed on the fiduced and shall consist of the examination of 4.5 and tests ofactor shall prepare a first article inspection report in accorda ordering document included in the contract (see 6.2.2).

4.4 Qualitspection. Quality conformance inspection shall consist of the e5 and the tests specified in 4.6.1 through 4.6.1.2.

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4.4.1 Sampling.

4.4.1.1 Inspection lot. For purposes of sampling, a lot shall consist of all electrodes presented at one time.

4.4.1.2 Sampling for examination and tests. A random sample of electrodes shall be selected in accordance with MIL-STD-105, general inspection level III, acceptance quality level (AQL) 1.5 percent.

4.5 Visual examination. Each of the electrodes selected in accordance with MIL-STD-105 shall be visually examined to verify compliance with all the requirements of this specification not involving tests. One or more defects specified in table I, of any electrode in the sample selected in accordance with 4.4.1.2, shall be cause for rejection of the electrode; and if the number of defective electrodes in any sample exceeds the acceptance number for that sample, the entire lot represented by the sample shall be rejected.

TABLE I. List of defects.

Critical	Defects
1.	Electrode cable and connector dimensions and weight not as specified.
2.	Electrode caps not securely affixed to electrode body.
3.	Electrode cable connections to electrode and to connectors defective.
4.	Adhesive tape on end cap closure missing or not covering vent hole.
5.	Electrolyte leakage from end cap closure, as evidenced by white deposits around end cap closure.
6.	Absorbent cotton liner inside end cap closure missing or dried out.
7.	Scratches, foreign material, smudges or cracks on any part of electrode assembly.
8.	U.S. standard type plug or pin tip connector bend or corroded.
9.	Electrode plastic body not of low actinic material.
10.	Preservation-packaging and marking not as specified.
11.	Instruction sheet missing or illegible.

4.6 Tests.

4.6.1 pH test. Each of the electrodes selected in accordance with MIL-STD-105 shall be tested using ASTM D 1293, method A to measure the pH of the reference buffer solutions as follows: The tartrate buffer shall be tested using the tetroxalate and phthalate buffers to calibrate the pH meter; the phosphate buffer shall be tested, using the phthalate and borax buffers to calibrate the pH meter; the sodium bicarbonate-sodium carbonate buffer shall be tested, using the borax and calcium hydroxide buffers to calibrate the pH meter. In these tests the combination electrode specified herein shall be used instead of the glass and reference electrode pair specified in ASTM D 1293, method A. The accuracy and precision criteria of 3.5.2 shall apply. None of the pH determinations shall require longer than 1 minute from time of immersion of the electrode in the test solution.

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4.6.1.1 Effect of temperature on pH. In order to ascertain that the combination electrode will respond properly to the change in solution pH with temperature, the combination electrode shall be used to measure the pH of the sodium bicarbonate-sodium carbonate buffer which shall be maintained at a temperature at least 10°C higher than that of the borax and calcium hydroxide buffers, which shall be used for calibrating the pH meter. The borax and calcium hydroxide buffers shall be in equilibrium with the ambient temperature. The pH measurement on the sodium bicarbonate-sodium carbonate buffer shall require no longer than 5 minutes after immersion of the combination electrode in the solution. The pH result obtained shall be compared with the tabulated pH values as specified in ASTM D 1293, or with the pH value interpolated therefrom. The precision and accuracy criteria as specified in 3.5.2 shall apply.

4.6.1.2 Effect of angular acceleration. The combination electrode shall be tested in accordance with 4.6.1 when the combination electrode and test sample are rolled about the normal position to an axis 60 degrees from the normal and back in a period of 12 seconds. The test shall be repeated about an axis 90 degrees removed. The accuracy and precision critieria of 3.5.2 shall apply.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation, packing and marking. pH electrodes, instruction sheets and data shall be preserved level A or C, packed level A, B or C as specified (see 6.2.1) and marked in accordance with MIL-E-17555. The sampling and inspection of the preservation, packing and marking shall be in accordance with the requirements of MIL-E-17555.

6. NOTES

6.1 Intended use. The combination pH electrode is intended to be used in conjunction with a digital indicating pH meter for the measurement of the pH of boiler water.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first article is required (see 3.1).
(c) Levels of preservation and packing required (see 5.1).

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.3	First article inspection report	DI-T-4902	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
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(Project 6630-N396)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-E-24636(SH)	ELECTRODE, pH, COMBINATION SEALED (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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